VALIDATION REPORT
OPERATIONAL QUALIFICATION
PROCESS CAPABILITY MODULE

JR Validated Environment — Process Capability Module v1.0

Field	Value
Document Number	JR-VR-CAP-001
Title	Validation Report — Operational Qualification, Process Capability Module
Validation Plan	JR-VP-CAP-001 v1.0
System	JR Validated Environment — Process Capability Module
Module Version	1.0
Document Version	1.0
Execution Date	2026-03-31
OQ Result	PASS — 45 / 45 tests passed, 0 failures, 0 deviations
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-03-24	Joep Rous	Initial OQ execution report. All 40 test cases passed. No deviations.
1.1	2026-03-31	Joep Rous	Report regenerated from evidence file. 45 / 45 tests passed.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-03-31
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document reports the execution and outcome of the Operational Qualification (OQ) test suite for the JR Validated Environment Process Capability Module, version 1.0. It provides objective evidence that all Cap scripts perform as specified in the OQ Validation Plan JR-VP-CAP-001 v1.0 under the defined test conditions.

This report records the test environment, execution results for all 45 test cases, and the formal OQ conclusion. Results are derived directly from the evidence file generated by admin_cap_oq.

2. Scope

2.1 In Scope

- All 3 R scripts in repos/cap/R/ (jrc_cap_normal, jrc_cap_nonnormal, jrc_cap_sixpack)
- Correct computation of Cp, Cpk, Pp, Ppk, Cpm for normally distributed data
- Correct computation of Pp and Ppk via percentile method for non-normal data
- Correct six-panel report generation by jrc_cap_sixpack
- Correct rejection of invalid inputs with informative error messages
- Bypass-protection verification (RENV_PATHS_ROOT check)
- PNG output to ~/Downloads/
- Numeric correctness: Cpk, Cp, Ppk verified against independently computed reference values for analytical datasets

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by JR-IQ-001)
- Performance qualification (PQ) with real production data
- Validation of R interpreter and third-party packages (ggplot2, grid)
- Statistical validation of the underlying methods (MR sigma estimation, percentile method) — method references are cited in JR-VP-CAP-001 Section 3

3. Reference Documents

Document ID	Title	Location
JR-VP-CAP-001 v1.0	Validation Plan — OQ, Process Capability Module	repos/cap/docs/cap_validation_plan.pdf
JR-VP-002 v1.0	Validation Plan — OQ, Community Script Suite v1.1.0	docs/oq_validation_plan.docx
JR-VP-001 v1.0	Validation Plan — Installation Qualification	docs/
JR-IQ-001	IQ Execution Evidence	docs/IQ_validation_20260311_205146.txt
OQ Evidence	Cap OQ Execution Evidence (pytest output + header)	cap_oq_execution_20260331T194153.txt

4. Test Environment

Component	Details

Execution date / time	2026-03-31T19:41:53
Host	Joeps-MacBook-Air.local
Operating System	15.7.3
R Version	4.5.3
Python Version	3.11.9
pytest Version	pytest 8.3.4
Test runner	repos/cap/admin_cap_oq
Test directory	repos/cap/oq/
Test data	repos/cap/oq/data/ (5 CSV fixtures)
Execution command	bash repos/cap/admin_cap_oq
OQ evidence file	cap_oq_execution_20260331T194153.txt

5. Test Results

All 45 test cases were executed on 2026-03-31. Results are derived from the OQ evidence file and reproduced below. The evidence file contains the complete pytest output including individual test IDs, pass/fail status, and timing.

Script	Tests Executed	Result
jrc_cap_normal	18	18/18 PASS
jrc_cap_nonnormal	13	13/13 PASS
jrc_cap_sixpack	14	14/14 PASS
TOTAL	45	45/45 PASS

5.1 jrc_cap_normal — 18/18 PASS

TC ID	Description	Expected	Result
TC-CAP-N-001	Valid input, both limits	exit 0, 'Capability' in output	**PASS**
TC-CAP-N-002	Cpk label present	exit 0, 'Cpk:' in output	**PASS**
TC-CAP-N-003	Cp label with both limits	exit 0, 'Cp:' in output	**PASS**
TC-CAP-N-004	Ppk label present	exit 0, 'Ppk:' in output	**PASS**
TC-CAP-N-005	Capable verdict — tight data	CAPABLE or EXCELLENT in output	**PASS**
TC-CAP-N-006	Marginal verdict — shifted data	MARGINAL or NOT CAPABLE in output	**PASS**
TC-CAP-N-007	PNG saved to ~/Downloads/	*_jrc_cap_normal.png in Downloads	**PASS**
TC-CAP-N-008	No arguments	exit non-zero, usage message	**PASS**
TC-CAP-N-009	File not found	exit non-zero	**PASS**
TC-CAP-N-010	n < 5	exit non-zero	**PASS**
TC-CAP-N-011	Non-numeric column	exit non-zero	**PASS**
TC-CAP-N-012	LSL > USL	exit non-zero	**PASS**
TC-CAP-N-013	Bypass protection	exit non-zero, RENV_PATHS_ROOT in output	**PASS**
TC-CAP-N-014	Cpk centred exact (=1.000 ± 0.005)	Cpk = 1.000 ± 0.005 (cap_cpk_1p000.csv)	**PASS**
TC-CAP-N-015	Cp centred exact (=1.000 ± 0.005)	Cp = 1.000 ± 0.005 (cap_cpk_1p000.csv)	**PASS**
TC-CAP-N-016	Cpk offset exact (=0.667 ± 0.005)	Cpk = 0.667 ± 0.005 (cap_cpk_0p667.csv)	**PASS**
TC-CAP-N-017	Cp location-independent (=1.000 ± 0.005)	Cp = 1.000 ± 0.005 (cap_cpk_0p667.csv)	**PASS**
TC-CAP-N-018	Ppk centred exact (=1.739 ± 0.020)	Ppk = 1.739 ± 0.020 (cap_cpk_1p000.csv)	**PASS**

5.2 jrc_cap_nonnormal — 13/13 PASS

TC ID	Description	Expected	Result
TC-CAP-NN-001	Valid non-normal input	exit 0, 'Capability' in output	**PASS**
TC-CAP-NN-002	Ppk label present	exit 0, 'Ppk' in output	**PASS**
TC-CAP-NN-003	P0.135 label present	exit 0, 'P0.135' in output	**PASS**
TC-CAP-NN-004	P99.865 label present	exit 0, 'P99.865' in output	**PASS**
TC-CAP-NN-005	Shapiro-Wilk reported	exit 0, 'Shapiro-Wilk' in output	**PASS**
TC-CAP-NN-006	No normal advisory for skewed	normality advisory absent	**PASS**
TC-CAP-NN-007	PNG saved to ~/Downloads/	*_jrc_cap_nonnormal.png in Downloads	**PASS**
TC-CAP-NN-008	No arguments	exit non-zero, usage message	**PASS**
TC-CAP-NN-009	File not found	exit non-zero	**PASS**
TC-CAP-NN-010	n < 5	exit non-zero	**PASS**
TC-CAP-NN-011	Non-numeric column	exit non-zero	**PASS**
TC-CAP-NN-012	LSL > USL	exit non-zero	**PASS**
TC-CAP-NN-013	Bypass protection	exit non-zero, RENV_PATHS_ROOT in output	**PASS**

5.3 jrc_cap_sixpack — 14/14 PASS

TC ID	Description	Expected	Result
TC-CAP-S-001	Valid input, both limits	exit 0, 'Sixpack' in output	**PASS**
TC-CAP-S-002	Cpk label present	exit 0, 'Cpk:' in output	**PASS**
TC-CAP-S-003	Ppk label present	exit 0, 'Ppk:' in output	**PASS**
TC-CAP-S-004	UCL and LCL reported	exit 0, 'UCL' and 'LCL' in output	**PASS**
TC-CAP-S-005	Shapiro-Wilk reported	exit 0, 'Shapiro-Wilk' in output	**PASS**
TC-CAP-S-006	SPC verdict present	exit 0, 'SPC:' in output	**PASS**
TC-CAP-S-007	Capability verdict present	exit 0, capability verdict word in output	**PASS**
TC-CAP-S-008	PNG saved to ~/Downloads/	*_jrc_cap_sixpack.png in Downloads	**PASS**
TC-CAP-S-009	No arguments	exit non-zero, usage message	**PASS**
TC-CAP-S-010	File not found	exit non-zero	**PASS**
TC-CAP-S-011	n < 5	exit non-zero	**PASS**
TC-CAP-S-012	Non-numeric column	exit non-zero	**PASS**
TC-CAP-S-013	LSL > USL	exit non-zero	**PASS**
TC-CAP-S-014	Bypass protection	exit non-zero, RENV_PATHS_ROOT in output	**PASS**

6. Deviations

No deviations were identified during OQ execution. All 45 test cases passed.

7. OQ Conclusion

The Operational Qualification of the JR Validated Environment Process Capability Module v1.0 is complete.

Result: PASS

All 45 automated test cases specified in JR-VP-CAP-001 v1.0 were executed on 2026-03-31 and 45 passed. Results are derived from the OQ evidence file listed below.

- jrc_cap_normal — Process Capability Analysis (Normal Distribution)

- jrc_cap_nonnormal — Process Capability Analysis (Non-Normal, Percentile Method)
- jrc_cap_sixpack — Process Capability Sixpack (Six-panel report)

Evidence of test execution is archived at: cap_oq_execution_20260331T194153.txt

The Process Capability module is released as version 1.0 of repos/cap/. Any subsequent change to the scripts, wrappers, or help files within repos/cap/ will require a change assessment and, if the change affects functionality covered by this OQ, re-execution of the affected test cases.